CHEMISTRY

1. Maximum I.E. and minimum I.E. of group 13 elements

(1) B, In

(2) B, Tl

(3) Al, In

(4) Al, Tl

Ans. (1)

Sol. Order of ionization energy : B > Tl > Ga > Al > In

2. Total number of electrons in chrominum (Z = 24) for which the value of azimuthal quantum number (l) is 1 and 2

Ans. (17)

Sol. Cr (24) : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

$l = 0 \rightarrow s$

$l = 1 \rightarrow p$

$l = 2 \rightarrow d$

Total number of electrons in p and d shell = 6 + 6 + 5

= 17

3. $\underset{150\text{Kg}}{CaCO_3} \xrightarrow[\Delta]{} CaO + CO_2$

Calculate the mass of CaO. ($CaCO_3$ is 75% Pure)

(1) 63 Kg

(2) 64 Kg

(3) 65 Kg

(4) None

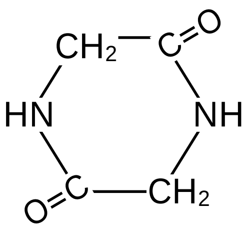
Ans. (1)

Sol. Mole of $CaCO_3$ = Mole of CaO

$$\frac{150}{100} \times \frac{75}{100} = \frac{\text{Mass of CaO}}{56}$$

Mass of CaO = 63 Kg

4. X is a peptide which is hydrolyse to 2 amino acids Y and Z. Y when reacted with HNO_2 gives lactic acid. Z when heated gives cyclic structure as below:



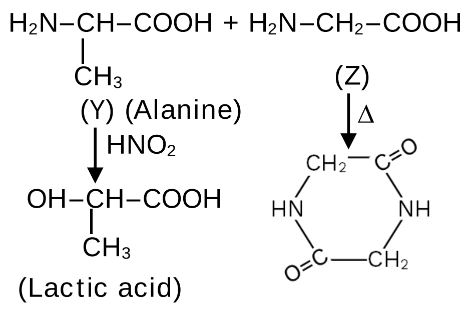
Y and Z respectively are:

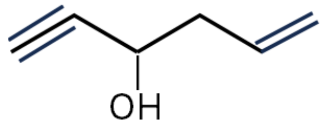
(1) Alanine and Lysine (2) Alanine and Glycine

(3) Glycine and Alanine (4) Valine and Glycine

Ans. (2)

Sol.



5. 

Correct IUPAC name of given Compound

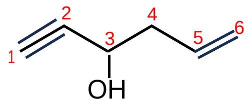
(I) hex-1-en-5-yne-4-ol

(II) hex-2-en-5-yne-4-ol

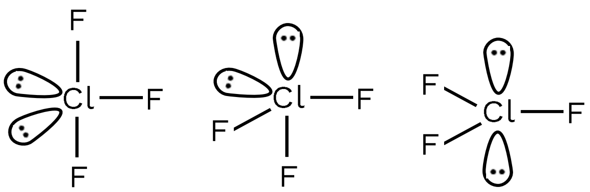
(III) hex-5-en-1-yn-3-ol

(IV) hex-1-en-5-yn-3-ol

Ans. (3)

Sol. 

6. Statement-I: Possible structure of ClF_3 are



Statement-II: In trigonal bipyramidal structure, lona pair lie on equatorial position.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are false

(2) Statement-I is false but Statement-II is true

(3) Both Statement I and Statement-II are true

(4) Statement-I is true but Statement-II is false

Ans. (2)

Sol. S-1 : As lone pair in TBP lies on equatorial position, it is incorrect.

S-2 : Correct.

7. Statement-1: Alcohol is prepared from alkyl halide in presence of aq.KOH by elimination.

Statement-2: Alkenes are prepared from alkyl halide with alc. KOH by β-elimination.

(1) Statement – 1 and Statement – 2 are correct

(2) Statement – 1 and Statement – 2 are incorrect

(3) Statement – 1 is true and Statement – 2 is false

(4) Statement – 1 is false and Statement – 2 is true

Ans. (4)

Sol. Statement-1: $R—X \xrightarrow{\text{aq.KOH}} R—OH$

Statement-2: $CH_3—CH_2—Br \xrightarrow{\text{alc.KOH}} CH_2{=}CH_2$

8. For a zero order reaction : $t_{1/2}$ = 1 hr and initial reactant concentration is 2M. Find time required for concentration to come from 0.5 to 0.25.

Ans. (15)

Sol. For Zero order reaction

$$t_{1/2} = \frac{C_0}{2K}$$

$$1 = \frac{2}{2K} \Rightarrow K = 1$$

$$t_{1/2} = \frac{0.5}{2 \times 1} = 0.25 \text{ hr} = 15\text{min}$$

= 15 min.

9. Order of number of unpaired electrons in the given complexes are:

(1) $[FeF_6]^{3-} < [Fe(CN)_6]^{3-} < [CoF_6]^{3-} < [Co(NH_3)_6]^{3+}$

(2) $[Co(NH_3)_6]^{3+} < [Fe(CN)_6]^{3-} < [CoF_6]^{3-} < [FeF_6]^{3-}$

(3) $[Co(NH_3)_6]^{3+} < [FeF_6]^{3-} < [CoF_6]^{3-} < [Fe(CN)_6]^{3-}$

(4) $[Co(NH_3)_6]^{3+} < [CoF_6]^{3-} < [FeF_6]^{3-} < [Fe(CN)_6]^{3-}$

Ans. (2)

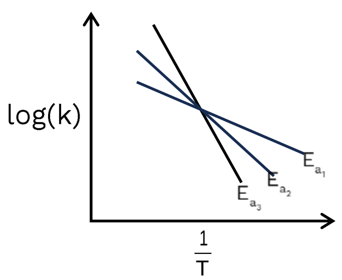
Sol. $[Fe(CN)_6]^{3-}$ $\rightarrow$ 1 unpaired electron

$[FeF_6]^{3-}$ $\rightarrow$ 5 unpaired electron

$[Co(NH_3)_6]^{3+}$ $\rightarrow$ 0 unpaired electron

$[CoF_6]^{3-}$ $\rightarrow$ 4 unpaired electron

10. Consider the following graph between log(K) and $\frac{1}{T}$.



(1) $E_{a_2} > E_{a_2} > E_{a_3}$ (2) $E_{a_3} > E_{a_2} > E_{a_1}$

(3) $E_{a_3} > E_{a_1} > E_{a_2}$ (4) $E_{a_1} > E_{a_3} > E_{a_2}$

Ans. (2)

Sol. $\log K = \log A - \frac{E_a}{2.303 \times R} \times \frac{1}{T}$

$$\text{Slope} = -\frac{E_a}{2.303 \times R}$$

More E_a means more negative slope.

11. Statement-1 Melting point elements of 14[th] group is greater than that of elements of 13[th] group.

Statment-2 I.E. of elements of 14[th] group is greater than that of 13[th] group.

(1) Both Statements 1 & 2 are correct

(2) Both Statements 1 & 2 are Incorrect

(3) Statement 1 is correct & Statement 2 is Incorrect

(4) Statement 1 is Incorrect & Statement 2 is correct

Ans. (1)

Sol. Factual (NCERT)

12. x mg of $Mg(OH)_{2(s)}$ is dissolved in water to make 1 lt solution of pH = 10. Calculate the value of x. (Nearest Integer)

Ans. (3)

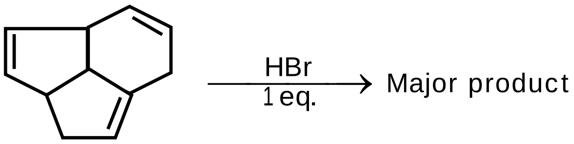
Sol. $Mg(OH)_2 \longrightarrow Mg^{2+} + 2OH^-$

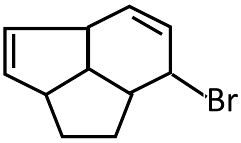
$pOH = 4 \Rightarrow [OH^-] = 10^{-4}$

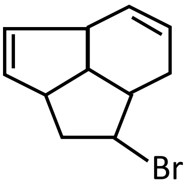
Mole of $Mg(OH)_2$ dissolved = $\frac{10^{-4}}{2}$

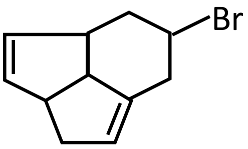
$\therefore$ Mass of $Mg(OH)_2$ dissolved

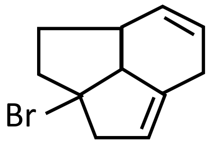
$= \frac{10^{-4}}{2} \times 58 \times 1000 = 2.9\text{mg} \simeq 3\text{mg}$

13. 

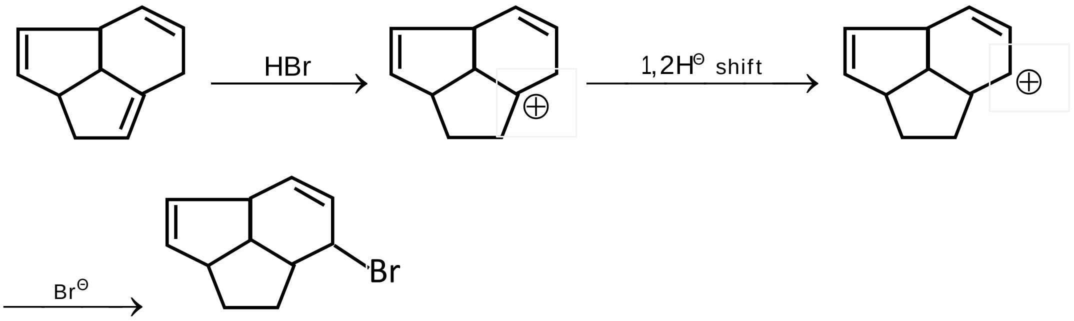
(1) 

(2) 

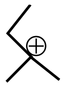
(3) 

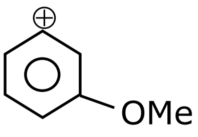
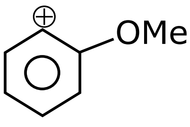
(4) 

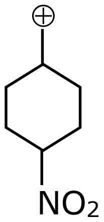
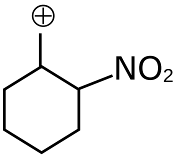
Ans. (1)

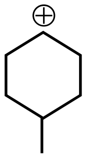
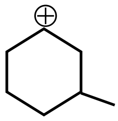
Sol. 

14. In which options I is more stable than II?

(A) (I)  (II) 

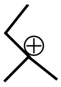
(B) (I)  (II) 

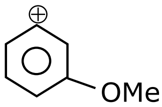
(C) (I)  (II) 

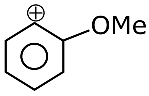
(D) (I)  (II) 

(1) A,D (2) A,B,C (3) B,C (4) All

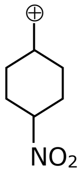
Ans. (2)

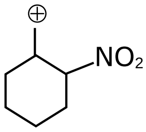
Sol. (A)  (9 αH) >  (8 αH)

(B) 

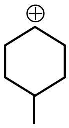
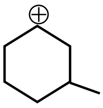
(–I) > 

(–I)

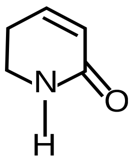
(C) 

(–I) > 

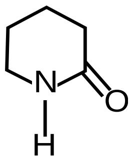
(–I)

(D)  (+I) <  (+I)

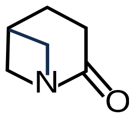
15. Correct basic strength of following compound is.



(A)



(B)



(C)

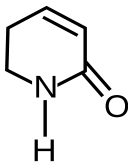
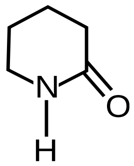
(1) C > A > B

(2) A > B > C

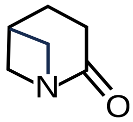
(3) B > A > C

(4) C > B > A

Ans. (1)

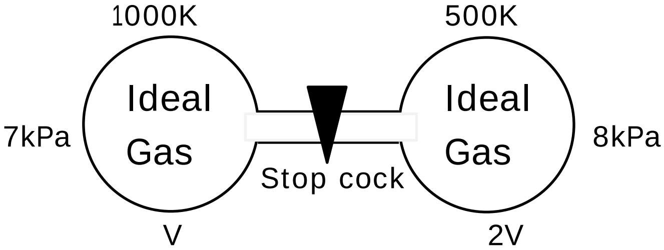
Sol.  

(cross conjugation) (lone pair of N is in conjugation with = O)



(conjugation not possible)

Hence C > A > B (Basic strength)

16. 

After opening stop cock for sufficiant time final temperature in the system is 600K. Find pressure of the gas.

(1) 4.44 kPa
(2) 6 kPa
(3) 7.8 kPa
(4) 3 kPa

Ans. (3)

Sol. $$\frac{7\times V}{1000R}+\frac{8\times 2V}{500R}=\frac{P_f\times(V+2V)}{600R}$$

$$\frac{7}{10}+\frac{16}{5}=\frac{P_f\times 3}{6}=\frac{P_f}{2}$$

$$\frac{7}{10}+\frac{16}{5}=\frac{P_f}{2}$$

$$P_f = \frac{7}{5} + \frac{32}{5} = 1.4 + 6.4 = 7.8 \text{ KPa.}$$

17.

List-I (Process)	List-II (Thermodynamic Parameter)
(P) Adiabatic	(1) $w = 0$
(Q) Isobaric	(2) $q = -w$
(R) Isochoric	(3) $q = 0$
(S) Isothermal	(4) $q = \Delta U + P\,\Delta V$

(1) P→3; Q→4; R→1; S→2
(2) P→1; Q→2; R→3; S→4
(3) P→2; Q→3; R→1; S→4
(4) P→4; Q→2; R→3; S→1

Ans. (1)

Sol. (P) Adiabatic → $q = 0$
(Q) Isobaric → $q = \Delta U + P\,\Delta V$
(R) Isochoric → $w = 0$
(S) Isothermal → $q = -w$

18. Given below are statements.

Statement I : The formula of cryoscopic constant is given as $K_f = \dfrac{MRT_f^2}{1000 \times \Delta S_{fusion}}$

Statement II : K_f of water is greater than benzene.

In light of the above statements choose the most appropriate option.

(1) Both Statements 1 & 2 are correct

(2) Both Statements 1 & 2 are Incorrect

(3) Statement 1 is correct & Statement 2 is Incorrect

(4) Statement 1 is Incorrect & Statement 2 is correct

Ans. (2)

Sol. $K_f = \dfrac{MRT_f^2}{1000 \times \Delta H_{fusion}}$

K_f of water (1.86) is less than benzene (more than 5).